

Xingjian Zhou

Xiamen University

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Education

B.S.	M.S.	Ph.D.	Joint Ph.D.
Southeast Univ. Nanjing, China 2016–2020	Xiamen Univ. Xiamen, China 2020–2022	Xiamen Univ. Xiamen, China 2022–present	Univ. of Trento Trento, Italy 2024–present

Major: Mathematics

Advisor (M.S. and Ph.D.): Prof. Chao Xia

Hosts (Joint Ph.D.): Profs. Virginia Agostiniani and Lorenzo Mazziere

Publications and Preprints

- [1] Chao Xia, Jiabin Yin, Xingjian Zhou, New monotonicity for p -capacitary functions in 3-manifolds with nonnegative scalar curvature, *Advances in Mathematics*, 440 (2024), 109526.
- [2] Jiabin Yin, Xingjian Zhou, Sharp upper bounds of p -capacity in convex cone and asymptotically flat half-space, *Letters in Mathematical Physics*, 116 (2026), Article 34.
- [3] Jiabin Yin, Xingjian Zhou, General monotone formula for homogeneous k -Hessian equation in the exterior domain and its applications, arXiv:2506.01434 (2025).
- [4] Chao Xia, Jiabin Yin, Xingjian Zhou, Mass- p -Capacity Inequalities in Asymptotically Flat Half-Spaces, arXiv:2605.31202 (2026).

Awards

National Scholarship of China, 2025.

Research Interests

Keywords: Potential theory; nonlinear elliptic and parabolic PDEs; inverse mean curvature flow; minimal surfaces; scalar curvature.

My research lies in geometric analysis, focusing on monotonicity formulas and geometric inequalities associated with elliptic equations arising in differential geometry. Current interests include scalar curvature problems, rigidity phenomena, and analytic aspects of the Penrose inequality.

Selected Talks and Posters

- Invited talk, Guangzhou Young Geometric Analysis Workshop, Guangzhou, China (2023).
- Invited talk, Southeast University, Nanjing, China (2024).
- Poster presentation, Topics in Geometric Analysis, Pisa, Italy (2025).
- Invited talk, Scuola Normale Superiore, Pisa, Italy (2025).
- Poster presentation, PRIN School on Geometry and Gravity, SISSA, Trieste, Italy (2025).
- Invited talk, Second Tianyuan Southeast Geometry Forum, Xiamen University, China (2026).
- Invited talk, University of Freiburg, Freiburg, Germany (2026).

Teaching Experience

- Teaching Assistant: Differential Geometry (Spring 2021); Differential Manifolds (Fall 2021, Fall 2022); Riemannian Geometry (Spring 2024).